

Quick Activity 1: School Supplies Calculator.

A student needs to buy notebooks and pens for their classes. The store sells notebooks for ₱30 each and pens for ₱15 each. Create a program to find the total cost of the items and the change the student will receive from their cash payment.

Decomposition

Assume the quantities bought and the payment amount are valid numbers.

INPUT	PROCESS	OUTPUT
Notebook quantity, pen quantity, payment amount	Compute total for notebooks, compute total for pens, add them together for final cost, subtract final cost from payment to get change	Final cost and change

Pseudocode

START

```
READ notebookQ, pen, payment
costNotebooks ← notebookQty × 30
costPens ← penQty × 15
finalCost ← costNotebooks + costPens
changeGiven ← cashPayment - finalCost
DISPLAY finalCost
DISPLAY changeGiven
```

END

Dry Run Calculations

Test Case 1

Input:

- Notebook = 3
- Pens = 2
- Payment = ₱150

STEP	VARIABLE	CALCULATIONS	SOLUTION
1	Notebook cost	3×30	₱90
2	Pen cost	2×15	₱30
3	Total cost	$90 + 30$	₱120
4	Change	$150 - 120$	₱30

Output

Final Cost = ₱120

Change Given = ₱30

Test Case 2

Input:

- Notebook = 5
- Pens = 4
- Payment = ₱500

STEP	VARIABLE	CALCULATIONS	SOLUTION
1	Notebook cost	5×30	₱150
2	Pen cost	4×15	₱60
3	Total cost	$150 + 60$	₱210
4	Change	$500 - 210$	₱290

Output

Final Cost = ₱210

Change Given = ₱290

